The use of a novel assessment protocol for the knee joint velocity proprioceptive sense to investigate motor learning abilities

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Introduction: Proprioception is a significant factor in balance, coordination, joint stability and movement acuity (1). Among proprioception's components, joint position sense (JPS) and movement detection have been mostly assessed whereas little is known about the joint velocity proprioceptive sense (2). Finding the joint angular velocity(-ies) that is more comprehensible by the human's brain, and therefore more accurately reproduced, could be used, among others, in motor learning rehabilitation protocols.

Research question: To investigate the knee angular velocities envelope inside which the joint can be moved with the most accuracy, depending on the task.

Methods: 48 subjects (23 men and 25 women) without knee pathology participated in the study (age 21.4 $\pm$3.85). Velocity replication (VR) was assessed in a spectrum of 5 different and randomly chosen low velocities (2°/s, 5°/s, 10°/s, 20°/s and 30°/s) by using concentric quadriceps contraction in an Isokinetic Dynamometer (Biodex System3 Pro). During the procedure the subjects were blindfolded, and the examiners were blind regarding the results.The passive demonstrations of each joint angular velocity were followed by active velocity replications. The number of passive demonstrations and active replications were adapted for each velocity in such a way, that each subject would stay almost the same amount of time, and therefore having the same effect or effort, in all of them.

Results: The knee angular velocities of 2°/s and 5°/s had the bigger mean percentage replication errors (68.2% and 29.0%) but the smallest mean errors in absolute value (1.4°/s for both the velocities). In the velocities of 10°/s, 20°/s and 30°/s the mean percentage replication errors were 26.3%, 26.1% and 29.0% respectively, while the mean errors in absolute value were 2.6°/s, 5.5°/s and 8.6°/s respectively.

Discussion: According to the present research, the knee joint can achieve a maximal precision of 1.4°/s angular velocity error, appear in joint velocities below 5°/s. Rehabilitation protocols require precision should focus is this kinematic envelope as, above this threshold, the angular velocity error increases gradually, as the joint velocity increases. For gross motor activities, where percentage joint angular velocity errors are more meaningful than the absolute error values, the kinematic envelope between 10°/s - 20°/s seem to be the ideal for motor learning tasks. According to our knowledge, this is the first attempt in the literature to investigate the knee angular velocity proprioception, and further investigation is needed on the velocity proprioceptive behavior of other joints, as well as any deviations in pathologies or trauma.

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The impact of quadriceps' fatigue on the proprioceptive perception of the knee joint position sense

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Introduction: Clinical importance of proprioception in feedback and feedforward control is an established knowledge (1). Among proprioceptors, muscle spindles are considered the most important. Muscle spindles are mainly detectors of speed stretch and length muscle whose information is necessary from CNS to compute the joint position and movement (2,3). Muscle fatigue produces neuromuscular deficits within the muscle, thus is a factor of joint injury. However, there are contradictory findings of muscle fatigue's impact on proprioception. Especially little is known about muscle fatigue on knee joint position sense (JPS) (4).

Research question: The investigation of the effect of quadriceps' fatigue on the knee joint position sense.

Methods: 54 healthy subjects (26 men and 28 women), without knee pathology, participated in the study (age 21.4 $\pm$2.88). JPS assessed in the knee joint within the ROM 90°-0° and the target angle defined in 45°. For all the measurements an Isokinetic Dynamometer (Biodex System 3 Pro) was used. The experiment conducted in three stages, in continuous time: a) a three times passive demonstration and passive replication of the targeted joint angle, b) the knee extensors fatigue procedure, and c) the repetition of the first stage with the knee extensors in fatigue state. Muscle fatigue was achieved by the standard protocol of continuous maximal isokinetic voluntary concentric contractions in 20°/sec, until three continuous maximal trials were under the 50% of the extensor's maximal peak torque. During the measurements the subjects were blindfolded.

Results: The average error deviation from the target angle before ($M = 4.202$, $SE = 0.23$) and after ($M = 4.691$, $SE = 0.31$) the fatigue protocol showed non-significant statistical difference $p = 0.127$. Partial analysis among the trials, before and after the fatigue, also supported the general results.

Discussion: Results showed that knee extensors' fatigue seems to not have impact on JPS. Probably the function of the knee joint proprioceptive receptors is not affected in the state of muscle fatigue and the brain keeps processing the information accurately. The results indicated that muscle spindles may not contribute as much as other joint proprioceptors in JPS.

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Comparison of spine structure, mobility, and competency in dentists with and without low back pain

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Introduction: Dental practices can cause musculoskeletal pain and dysfunction due to cumulative microtrauma and inappropriate working positions (1). The prevalence of musculoskeletal pain in dentists was declared between 64% and 93% (2). The prevalence of low back pain in dentists was reported as 47.6% as the most common musculoskeletal dysfunction (3,4).

Research question: Is there a difference in spine structure, mobility, and competency of dentists with low back pain compared to those without low back pain?

Methods: In the study, 65 dentists with low back pain (40 females, 25 males, age: 25.57 ± 2.83 years, weight: 67.64 ± 13.20 kg, height: 171.72 ± 8.46 cm, BMI, 22.75 ± 3.25 kg/m2) and 57 pain-free matched control group (30 females, 27 males, age: 26.36 ± 3.94 years, weight: 69.05 ± 13.00 kg, height: 170.53 ± 7.78 cm, BMI, 23.64 ± 3.52 kg/m2) were included. Spine structure, mobility, and competency in the sagittal and frontal planes were evaluated with the Valedo®Shape device (Idiag, Fehraltorf, Switzerland). Parameters were obtained in degrees: thoracic, lumbar, sacral/hip angle, and trunk inclination angle (angle between straight line from T1 to S1 and vertical line). After the spinous processes of the spine were marked as reference points, the Valedo®Shape device was moved down by the evaluator over all the spinous processes starting from the C7 spinous process to approximately the S3 spinous process. The response of the spine to loading was evaluated using weight for competency measurement. After measuring before the weight, the participant was asked to wait for 30 seconds with the weights in hand, and the measurement was repeated (6). The normality distributions of the data were determined by the Shapiro-Wilk test. In the comparison of the data, the independent sample t-test was used in those with normal distribution, and the Mann-Whitney U test was used for those that were not normally distributed.

Results: In patients with low back pain, in the sagittal plane, the inclination angle decreased ($p = 0.045$), there was a shift in the sacral angle with loading ($p = 0.037$). In the sagittal and frontal planes, there was no significant difference in thoracic region angles ($p = 0.292$;0.074) and in the lumbar region angles ($p = 0.369$; $p = 0.781$).

Discussion: In participants with low back pain, the angle of inclination decreased in the sagittal plane and a lateral shift response occurred in the sacrum with loading. It is known that the angles of the lumbopelvic region are directly related to the curvature of the spine and compensatory mechanisms against spinal deformities in the sagittal plane in this region (7). In a previous study, a shift in the inclination angle was reported by dentists with low back pain (8). The changes in the spine in dentists may be seen in the occurrence of low back pain. This should be considered for the assessment and treatment of low back pain.

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Investigation of the relationship between measurement of scapular asymmetry and working posture in dentists

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Introduction: Dentists are at risk due to adverse conditions they are exposed to, such as improper working posture, repetitive movements, long-term static positions, excessive effort with frequent use of small muscles, tight grip of materials, using vibrating instruments, and holding their arms high for long periods of time (1).

Research question: Does the scapular asymmetry distance increase as the working posture of dentists worsens?